Ex. $f(x) = xe^{x^2}$. Find Maclarin Series for $f(x)$

We know from prev. examples that $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

So then $e^{x^2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$ ← $(x^n)^2 = x^{2n}$

↳ $xe^{x^2} = \sum_{n=0}^{\infty} \frac{x^{2n} x}{n!} = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{n!}$

you don't need to plug in values for a function, you can start from a simpler function

$f(x) = xe^{x^2}$. Find Taylor Series for $f(x)$ centered at $a = 1$ → First 3 terms

Now we can't build off prev. example since center has changed

| $f^{(n)}(x)$ | $f^{(n)}(a) = f^{(n)}(1)$ |
|---|---|
| $f(x) = xe^{x^2}$ | $f(1) = e$ |
| $f'(x) = xe^{x^2} \cdot 2x + e^{x^2} = e^{x^2}(2x^2+1)$ | $f'(1) = 3e$ |
| $f''(x) = e^{x^2} \cdot 2x(2x^2+1) + 4x \cdot e^{x^2} = e^{x^2}(4x^3+6x)$ | $f''(1) = 10e$ |
| | $\vdots$ |

Taylor series: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$

$= f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 + \ldots$

↳ $e + 3e(x-1) + 5e(x-1)^2 + \ldots$

If a Taylor series for $f(x)$ about $x = a$ is given as

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k,$$

then the $n$th degree Taylor polynomial approximation of $f(x)$ about $x = a$ is

↳ of degree $k$

$$T_k(x) = \sum_{n=0}^{k} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Degree refers to highest power, not # of terms

How well $T_k(x)$ approximates $f(x)$ depends on $x$

**Important Taylor/Maclaurin series:**

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

all centered at 0

**Example:** Find a Taylor series about $x = 0$ for $f(x) = xe^{x^2}$.

**Example:** Find the sum of the series $\sum_{n=0}^{\infty} (-1)^n \frac{\pi^{2n}}{6^{2n}(2n)!}$

**Example:**

(a) State the Taylor series for $f(x) = e^x$ about $x = 0$.

$$f(x) = e^x = \sum_{n\geq 0}^{\infty} \frac{x^n}{n!}$$

(b) Find the degree 1 Taylor polynomial approximation of $f(x)$ about $x = 0$.

1 st degree = linear

$$T_1(x) = 1 + x$$

(c) Find the degree 3 Taylor polynomial approximation of $f(x)$ about $x = 0$.

degree 3 = cubic

$$T_3(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$$

How well $T_k(x)$ approximates $f(x)$ depends on $x$

If $x = 0$: $e^0 = 1$, $T_1(0) = 1$, $T_3(0) = 1$

If $x = 1$: $e^1 = e \approx 2.71$, $T_1(1) = 2$, $T_3(1) = 2 + \frac{2}{3}$

**Taylor remainder inequality:**

Suppose the Taylor series for $f(x)$ is given by:

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

and the $n$th degree Taylor polynomial approximate is given by:

$$f(x) \approx T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

If $|f^{(n+1)}(x)| \leq M$ for $|x-a| \leq d$, then

$$|R_n(x)| = |f(x) - T_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}, \quad \text{for } |x-a| \leq R$$

Remainder:

$$R_k(x) = f(x) - T_k(x)$$
$$= \sum_{n=k+1}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Bound:

If $|f^{(k+1)}(x)| \leq M$ for $|x-a| < \boxed{R}$ radius of convergence

a constant that is a bound

commonly: $|\cos(x)| \leq 1$, $|\sin(x)| \leq 1$
So M would be 1 in both cases

Then $|R_k(x)| \leq \underbrace{\frac{M}{(k+1)!}|x-a|^{k+1}}$

Biggest term in $R_k(x)$?

$$T_k(x) = \sum_{n=0}^{k} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$

$\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$

**Example:** Consider $f(x) = \cos x$

(a) Find a 4th degree Taylor polynomial approximation to $f(x)$ about $a = \frac{\pi}{3}$.

$K = 4$

| $f^{(n)}(x)$ | $f^{(n)}\left(\frac{\pi}{3}\right)$ |
|---|---|
| $f(x) = \cos(x)$ | $\frac{1}{2}$ |
| $f'(x) = -\sin(x)$ | $-\frac{\sqrt{3}}{2}$ |
| $f''(x) = -\cos(x)$ | $-\frac{1}{2}$ |
| $f'''(x) = \sin(x)$ | $\frac{\sqrt{3}}{2}$ |
| $f''''(x) = \cos(x)$ | $\frac{1}{2}$ |
| $f'''''(x) = -\sin(x)$ | $-\frac{\sqrt{3}}{2}$ |

$$T_4(x) = \underbrace{\frac{1}{2}}_{\frac{f(\frac{\pi}{3})}{0!}(x-\frac{\pi}{3})^0} \underbrace{- \frac{\sqrt{3}}{2}\left(x - \frac{\pi}{3}\right)}_{\frac{f'(\frac{\pi}{3})}{1!}(x-\frac{\pi}{3})^1} \underbrace{- \frac{1}{4}\left(x - \frac{\pi}{3}\right)^2}_{\frac{f''(\frac{\pi}{3})}{2!}(x-\frac{\pi}{3})^2} + \underbrace{\frac{\sqrt{3}}{12}\left(x - \frac{\pi}{3}\right)^3}_{\frac{f'''(\frac{\pi}{3})}{3!}(x-\frac{\pi}{3})^3} + \underbrace{\frac{1}{48}\left(x - \frac{\pi}{3}\right)^4}_{\frac{f''''(\frac{\pi}{3})}{4!}(x-\frac{\pi}{3})^4}$$

(b) How good is the approximation when $0 \le x \le \frac{2\pi}{3}$?

For $|\underbrace{f^{(5)}(x)}_{-\sin(x)}| \le M$ for $0 \le x \le \frac{2\pi}{3}$, then $|R_5(x)| \le \frac{M}{5!}\left|x - \frac{\pi}{3}\right|^5$

$|-\sin(x)| \le 1 = M$

$$|R_5(x)| \le \frac{1}{5!}\left|x - \frac{\pi}{3}\right|^5 \quad \text{for } 0 \le x \le \frac{2\pi}{3}$$

$$\le \underbrace{\frac{1}{5}\left(\frac{\pi}{3}\right)^5}_{g(x) = \frac{1}{5!}|x - \frac{\pi}{3}|^5} \approx 0.01\ 01$$

$g(0) = \frac{1}{5!}\left(\frac{\pi}{3}\right)^5$

$g\left(\frac{2\pi}{3}\right) = \frac{1}{5!}\left(\frac{\pi}{3}\right)^5$

Now we need to Maximize $|R_5(x)|$ * why?

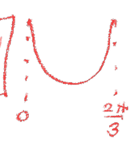
$g(x) = \frac{1}{5!}\left|x - \frac{\pi}{3}\right|^5$

via chain rule

$g'(x) = \frac{1}{5!}\, 5\left|x - \frac{\pi}{3}\right|^4 (\pm 1)$

If $g'(x) = 0$, $x = \frac{\pi}{3}$ (local min b/c $g(\frac{\pi}{3}) = 0$)

So to find the local max we need to test the endpoints - only 1 critical point found within interval, and it's a min




$0 \le x \le \frac{2\pi}{3}$

$g(0) = \frac{1}{5!}\left(\frac{\pi}{3}\right)^5$

$g\left(\frac{2\pi}{3}\right) = \frac{1}{5!}\left(\frac{\pi}{3}\right)^5$

biggest it can be is at endpoints